HAILEMICHAEL TESFAYE MEKURIA

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PROFESSIONAL PROFILE

Certified Agentic AI Engineer and Full-Stack Developer specializing in multi-agent orchestration (LangGraph, FastMCP), spatial digital twins, and autonomous resilient systems. Proven track record leading campus AI initiatives, engineering edge computer vision workflows (Meta SAM 3.1), and integrating serverless GPU clusters (Modal, Baseten) with strict constitutional safety verification. Passionate about architecting low-latency, sovereign AI infrastructure for Global South urban resilience.

EDUCATION

Bachelor of Science in Information Technology **Oct 2022 – Aug 2026 (Expected)**

Wollo University, Kombolcha Institute of Technology | Dessie, Ethiopia

Relevant coursework: Artificial Intelligence, Distributed Systems, Data Structures & Algorithms, Database Management Systems, Computer Networks & Security, Software Engineering, Critical Infrastructure Protection (CIP).

WORK & LEADERSHIP EXPERIENCE

Campus Lead & Technical Ambassador **Jul 2026 – Present**

The Udara Project / NSK AI · Wollo University | Ethiopia

- Spearheaded technical development programs empowering 150+ African campus students with hands-on Agentic AI and multi-agent cyclic DAG architectures using LangGraph.
- Represented Wollo University in international technical lead selection, coordinating high-impact workshops bridging academic computer science with real-world agent orchestration.
- Collaborated with global engineering teams across Zerra Build and NSK AI to validate hands-on agentic workflows and sovereign data isolation protocols.

Full-Stack & Agentic AI Developer **Jan 2025 – Present**

Independent Engineering & Global Hackathons | Addis Ababa, Ethiopia

- Engineered full-stack reactive applications using React, TypeScript, Node.js, Express, and PostgreSQL/MySQL with sub-150ms roundtrip response times.
- Architected client-side PII/PHI sanitizers stripping sensitive biometric, national ID, and telecom credentials before LLM dispatch, guaranteeing zero data egress.
- Integrated Model Context Protocol (FastMCP) tool servers connecting autonomous agents with spatial sensors, physical actuators, and structured JSON schemas.

FEATURED ENGINEERING PROJECTS

NEXUS: The Living City – Autonomous Multi-Agent Digital Twin **Oct 2026 – Present**

Meta Global AI Developer Hackathon Submission | Track: Build for Your World

- Architected an autonomous 11-agent cognitive digital twin for Addis Ababa crisis management using Meta Llama 3.3 70B arbitration and Meta SAM 3.1 drone computer vision.
- Deployed distributed GPU inference pipelines across Modal and Baseten clusters (Muse Spark Open), achieving sub-120ms multi-agent dispute resolution with Llama-Guard 3 safety floors.
- Engineered genuine municipal physical actuation triggers (8,000L/min flood pumps, Gotera smart traffic green waves, and Tikur Anbessa ICU reservations) with bilingual Amharic/English voice dispatch.

NetsaGuard AI – Multilingual Privacy-Preserving Content Safety Hub **2026**

LangGraph Cyclic StateMachine, Groq LPU, ChromaDB Vector Store, FastMCP

- Developed an Africa-first agentic moderation engine using LangGraph cyclic state machines and ChromaDB cosine vector retrieval for high-speed local language filtering.
- Constructed Human-In-The-Loop (HITL) cryptographic approval gates with SHA-256 tamper-proof audit manifests for municipal and enterprise governance.

Evangadi Forum – Full-Stack Community Q&A Platform **2025**

React, Node.js, Express, MySQL, JWT Authentication

- Designed an interactive full-stack discussion platform featuring secure JWT authentication, relational MySQL schema optimization, and responsive mobile-first UI.

TECHNICAL SKILLS

- Agentic AI & LLMs: LangGraph, Cyclic DAGs, Model Context Protocol (FastMCP), Meta Llama 3.3 70B, Llama-Guard 3, Meta SAM 3.1, ChromaDB, RAG, Prompt Engineering, Groq LPU.
- Programming Languages: TypeScript, JavaScript (ES6+), Python, SQL, C++, HTML5, CSS3/Tailwind.
- Frontend & 3D Spatial: React 19, Next.js, Redux Toolkit, TailwindCSS, Mapbox GL JS, Three.js, Lucide.
- Backend & Databases: Node.js, Express.js, FastAPI, RESTful APIs, WebSockets, PostgreSQL, MySQL.
- Infrastructure & Cloud: Modal Serverless GPUs, Baseten Engine, Docker, Vercel, Git/GitHub.
- Security & Specializations: Critical Infrastructure Protection (CIP), Human-In-The-Loop (HITL), Sovereign Privacy Filtering.

CERTIFICATIONS & VERIFIED CREDENTIALS

October 2026	OPSWAT Academy: Introduction to Critical Infrastructure Protection (ICIP) (ID: ohD12ZSzzA)
August 2026	NSK AI / Zerra Build Global Cohort: Certified Agentic Engineering Curriculum (ID: Verified)
July 2026	TheUdaraProject (Crew COREMIND): The Udara Project - Agentic AI Solutions Workflow (ID: carluwrn40ige14179lomim